

IMU Preintegration for Visual-Inertial Odometry Pose Estimation

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Abstract: In this paper, based on Inertial Measurement Unit (IMU) preintegration, a pose estimation algorithm for Visual-Inertial Odometry (VIO) system has been investigated. Firstly, we developed the kinematic equation of acceleration and angular velocity of IMU. To avoid the re-integration caused by the change of the system states, the IMU measurements are preintegrated on the Riemannian manifold of rotation. Furthermore, we studied the propagation characteristic of noise and derived the state transition equation of the noise. So the influence of noise on pose estimation can be dramatically reduced. Compared with the famous Okvis algorithm, experimental results on two public datasets of EuRoC show that the proposed IMU preintegration algorithm achieves good estimation median of translation error with 0.14 m, which is better than Okvis with 0.24 m.

Key Words: IMU Preintegration, Manifold, Kinematic Equation, VIO, Noise Propagation

1 Introduction

Visual Odometry (VO) and Simultaneous Localization And Mapping (SLAM) have become an important part of autonomous navigation research because they provide a convenient and reliable alternative to classic robot/vehicle localisation solutions [1]. Compared to established environment-perceiving sensors, like laser scanners, cameras have the advantage that they are lightweight, can detect most solid materials, and yield readings with comparably high frequency [2]. Hence, they are essential in navigation, motion planning, and obstacle avoidance. However, in typical applications the camera field of view is limited. Close obstacles occlude large parts of the scene, structureless surfaces often lack visual cues, and repetitive textures complicate finding correspondences. Using inertial sensor can help in mitigating these effects. A sensor setup comprising a camera and an IMU is very suitable for navigation, but imposes great research challenges. Fusing inertial and visual cues can be very beneficial as they provide complementary information, aiding the robustness of the estimation processes [3].

Related work on visual-inertial odometry can be mainly divided into two kinds, based on the filtering method and based on nonlinear optimization method. Filtering algorithms enable efficient estimation by restricting the inference process to the latest state of the system. Pang presented a VIO system based on Multi-State Constraint Kalman Filter (MSCKF), which benefits from low computation load when compared to optimization-based method [4]. Sirtkaya proposed a loosely coupled indirect feedback Kalman filter integration for visual odometry and inertial navigation system that is based on error propagation model. A practical measurement model approach is proposed for the delta position and attitude change measurements [5].

Optimization methods estimate the history of states (cam-

era trajectory and 3D landmarks), by solving a large nonlinear optimization problem [6]. Hong proposed a novel VIO algorithm which directly optimizes the camera poses with noisy IMU data and visual feature locations [7]. Kasyanov presented a keyframe-based approach to visual-inertial SLAM, which is based on a real-time capable VIO method that provides locally consistent trajectory and map estimates. The method achieve global consistency in the estimate through online loop-closing and non-linear optimization [8]. The above literatures use quaternion to represent the rotation process for optimization. In nonlinear optimization, the description of rotation with Lie algebra is a nature way and it is convenient for optimization computation, such as in g2o optimization framework [9]. Motivated from the work of [10], the contribution of this paper is to propose a pose estimation algorithm based on the IMU pre-integration in manifold, which uses Lie algebra to represent the rotation process to optimize. We consider the influence of noise and bias during the estimation. Finally, some experiments are given to demonstrate the effectiveness of the proposed approach.

The remainder of this paper is organized as follows. Section 2 introduces the basic knowledge of manifold. The IMU preintegration algorithm is presented in Section 3. The experimental results on the public datasets of the algorithm are given in Section 5. Finally, the conclusion of our work is given in Section 6.

2 Preliminaries

This section mainly introduces the related knowledge of Lie group $SO(3)$ and Lie algebra $\mathfrak{so}(3)$.

The rotation group $SO(3)$ is a Riemannian manifold defined on a 3-dimensional Euclidean space and can be used to describe the rotation in 3-D space.

All three-dimensional rotation matrix constitutes Lie group $SO(3)$:

$$SO(3) \doteq \{ \mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^T \mathbf{R} = \mathbf{I}, \det(\mathbf{R}) = 1 \}.$$

We define the symbol $(\cdot)^\wedge$ to represent the skew-symmetric matrix corresponding to the rotation vector, that

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is:

$$\omega^\wedge \doteq \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}.$$

Lie algebra $\mathfrak{so}(3)$ is a set of three-dimensional skew-symmetric matrices, which reflects the derivative nature of the corresponding rotation matrix:

$$\mathfrak{so}(3) \doteq \{\phi^\wedge \mid \phi \in \mathbb{R}^3, \phi^\wedge \in \mathbb{R}^{3 \times 3}\}.$$

Lie algebra is the tangent space on the manifold $SO(3)$. The three-dimensional rotation vector ϕ , skew-symmetric matrix $\phi^\wedge$ and rotation matrix $\mathbf{R}$ are one-to-one correspondence. They satisfy the following properties:

$$\begin{aligned} \mathbf{R} &= \exp(\phi^\wedge) \\ \dot{\mathbf{R}} &= \phi^\wedge \mathbf{R}. \end{aligned}$$

In this paper, we define the following operators to associate the three-dimensional rotation vector ϕ with the corresponding rotation matrix $\mathbf{R}$:

$$\begin{aligned} \text{Exp}(\phi) &= \exp(\phi^\wedge) = \mathbf{R} \\ \text{Log}(\mathbf{R}) &= \log(\mathbf{R})^\vee = \phi. \end{aligned} \quad (1)$$

The symbol $(\cdot)^\vee$ map the skew-symmetric matrix to the rotation vector.

The first-order Taylor expansion of exponential map of Lie algebraic can be approximated as:

$$\text{Exp}(\phi) \approx \mathbf{I} + \phi^\wedge. \quad (2)$$

By adding a small perturbation $\Delta\phi$ on the Lie algebra ϕ , the perturbation model can be approximated as an additional multiplication of a Jacobian matrix on Lie group:

$$\text{Exp}(\phi + \Delta\phi) \approx \text{Exp}(\phi) \text{Exp}(\mathbf{J}_r(\phi) \Delta\phi). \quad (3)$$

Where $\mathbf{J}_r(\phi)$ relates additive increments in the tangent space to multiplicative increments applied on the right-hand-side of $SO(3)$. Let's define $\phi = \mathbf{a}\theta$, θ is the rotation angle and $\mathbf{a}$ is the unit rotation vector that has the same direction as ϕ . we get:

$$\mathbf{J}_r(\phi) = \frac{\sin \theta}{\theta} \mathbf{I} + (1 - \frac{\sin \theta}{\theta}) \mathbf{a} \mathbf{a}^\top - \frac{1 - \cos \theta}{\theta} \mathbf{a}^\wedge \quad (4)$$

3 IMU Preintegration

In this section, after giving the model of IMU, we analyze the kinematic equations of the VIO system, and obtain the IMU preintegral equation. Finally, we consider the impact of measuring noise and bias on the system.

3.1 IMU Model

The IMU sensor used in this paper is a six-axis sensor which provide acceleration and angular velocity information. Considering the measurement noise and bias, the IMU measurement model is constructed as follows:

$$\begin{aligned} {}_B\hat{\omega}(t) &= {}_B\omega(t) + \mathbf{b}_g(t) + \boldsymbol{\eta}_g(t) \\ {}_B\hat{\mathbf{a}}(t) &= \mathbf{R}_{WB}^\top(t)({}_W\mathbf{a}(t) + {}_W\mathbf{g}) + \mathbf{b}_a(t) + \boldsymbol{\eta}_a(t). \end{aligned} \quad (5)$$

For the sake of convenience, in this paper, W represents the world coordinate frame and B represents the IMU coordinate frame.

From equation (5), we know that the IMU measurements are affected by the bias $\mathbf{b}_g(t)$ and $\mathbf{b}_a(t)$, the white noise $\boldsymbol{\eta}_g(t)$ and $\boldsymbol{\eta}_a$ and the gravity vector ${}_W\mathbf{g}$. We model the white noise as Gaussian distribution, that is $\boldsymbol{\eta}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2)$ and $\boldsymbol{\eta}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2)$. The bias are modeled as random walk process, and their differential is Gaussian white noise, that is $\boldsymbol{\eta}_{b_a} \sim \mathcal{N}(\mathbf{0}, \sigma_{b_a}^2)$ and $\boldsymbol{\eta}_{b_g} \sim \mathcal{N}(\mathbf{0}, \sigma_{b_g}^2)$.

3.2 Kinematic Equation

For the IMU system, the kinematic differential equation is given by:

$$\begin{aligned} \dot{\mathbf{R}}_{WB} &= \mathbf{R}_{WB} \omega^\wedge \\ {}_W\dot{\mathbf{v}} &= {}_W\mathbf{a} \\ {}_W\dot{\mathbf{p}} &= {}_W\mathbf{v}. \end{aligned} \quad (6)$$

Assuming that ${}_B\omega$, ${}_B\mathbf{a}$, ${}_B\mathbf{v}$ remains constant for a short period of time $t \sim t + \Delta t$, the states at time $t + \Delta t$ are obtained by integrating Eq.(6):

$$\begin{aligned} \mathbf{R}_{WB}(t + \Delta t) &= \mathbf{R}_{WB}(t) \text{Exp}({}_B\omega(t) \Delta t) \\ {}_W\mathbf{v}(t + \Delta t) &= {}_W\mathbf{v}(t) + {}_W\mathbf{a}(t) \Delta t \\ {}_W\mathbf{p}(t + \Delta t) &= {}_W\mathbf{p}(t) + {}_W\mathbf{v}(t) \Delta t + \frac{1}{2} {}_W\mathbf{a}(t) \Delta t^2. \end{aligned} \quad (7)$$

The discrete kinematics of the system can be obtained by discretizing Eq.(7):

$$\begin{aligned} \mathbf{R}_j &= \mathbf{R}_i \prod_{k=i}^{j-1} \text{Exp}(\omega_k \Delta t) \\ \mathbf{v}_j &= \mathbf{v}_i + \sum_{k=i}^{j-1} \mathbf{a}_k \Delta t \\ \mathbf{p}_j &= \mathbf{p}_i + \sum_{k=i}^{j-1} \mathbf{v}_k \Delta t + \frac{1}{2} \sum_{k=i}^{j-1} \mathbf{a}_k \Delta t^2, \end{aligned} \quad (8)$$

where,

$$\begin{aligned} \mathbf{R}_i &\doteq \mathbf{R}_{WB}(i), & \mathbf{v}_i &\doteq {}_W\mathbf{v}_i, \\ \mathbf{p}_i &\doteq {}_W\mathbf{p}_i, & \mathbf{a}_i &\doteq {}_W\mathbf{a}_i, & \mathbf{w}_i &\doteq {}_B\mathbf{w}_i. \end{aligned}$$

By substituting the equation (5) into the equation (8), we can get the system states represented by the IMU measurements and noise:

$$\begin{aligned} \mathbf{R}_j &= \mathbf{R}_i \prod_{k=i}^{j-1} \text{Exp}((\hat{\omega}_k - \mathbf{b}_{gk} - \boldsymbol{\eta}_{gk}) \Delta t) \\ \mathbf{v}_j &= \mathbf{v}_i + \sum_{k=i}^{j-1} (\mathbf{R}_k ({}_B\hat{\mathbf{a}}_k - \mathbf{b}_{ak} - \boldsymbol{\eta}_{ak}) - {}_W\mathbf{g}) \Delta t \\ \mathbf{p}_j &= \mathbf{p}_i + \sum_{k=i}^{j-1} \left(\mathbf{v}_k \Delta t + \frac{1}{2} (\mathbf{R}_k ({}_B\hat{\mathbf{a}}_k - \mathbf{b}_{ak} - \boldsymbol{\eta}_{ak}) - {}_W\mathbf{g}) \Delta t^2 \right). \end{aligned} \quad (9)$$

3.3 IMU preintegration

From the equation (9), the current states of the system can be computed quickly according to the original states and the IMU input measurement. However, in the SLAM system, the original states of the system are optimized at every

sampling frame. The re-integration of IMU measurement is needed after pose optimization. To avoid the re-propagation, we adopt preintegration algorithm.

We can only preintegrate the parts which are related to the IMU measurement as follows:

$$\begin{aligned} \mathbf{R}_j &= \mathbf{R}_i \Delta \mathbf{R}_{ij} \\ \mathbf{v}_j &= \mathbf{v}_i - \mathbf{w} \mathbf{g} \Delta t_{ij} + \mathbf{R}_i \Delta \mathbf{v}_{ij} \\ \mathbf{p}_j &= \mathbf{p}_i + \mathbf{v}_i \Delta t_{ij} - \frac{1}{2} \mathbf{w} \mathbf{g} \Delta t_{ij}^2 + \mathbf{R}_i \Delta \mathbf{p}_{ij}, \end{aligned} \quad (10)$$

where,

$$\begin{aligned} \Delta \mathbf{R}_{ij} &\doteq \prod_{k=i}^{j-1} \text{Exp}((\hat{\omega}_k - \mathbf{b}_{gk} - \boldsymbol{\eta}_{gk}) \Delta t) \\ \Delta \mathbf{v}_{ij} &\doteq \sum_{k=i}^{j-1} \Delta \mathbf{R}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ak} - \boldsymbol{\eta}_{ak}) \Delta t \\ \Delta \mathbf{p}_{ij} &\doteq \sum_{k=i}^{j-1} \frac{3}{2} \Delta \mathbf{R}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ak} - \boldsymbol{\eta}_{ak}) \Delta t^2. \end{aligned}$$

The equation (10) shows that the preintegration of the IMU measurement and the states of the system at time i are independent of each other. We can first perform IMU preintegration to get $\Delta \mathbf{R}_{ij}$, $\Delta \mathbf{v}_{ij}$ and $\Delta \mathbf{p}_{ij}$. When the system states at time i are changed, we can quickly compute the estimated states at time j by using the results of IMU preintegration.

3.4 Noise Propagation

In the equation (10), we know the preintegration variable $\Delta \mathbf{R}_{ij}$, $\Delta \mathbf{v}_{ij}$ and $\Delta \mathbf{p}_{ij}$ are independent of the initial states of the system. However, they are affected by noise and bias. We assume that the bias remain constant between two keyframes, and then analyze the effect of white noise:

$$\begin{aligned} \mathbf{b}_{gi} &= \mathbf{b}_{gi+1} = \dots = \mathbf{b}_{gj-1} \\ \mathbf{b}_{ai} &= \mathbf{b}_{ai+1} = \dots = \mathbf{b}_{aj-1}. \end{aligned}$$

Firstly, we consider the rotation IMU preintegration variable $\Delta \mathbf{R}_{ij}$. Using equation (3) and (4), we isolate the white noise from $\Delta \mathbf{R}_{ij}$:

$$\begin{aligned} \Delta \mathbf{R}_{ij} &\approx \prod_{k=i}^{j-1} [\text{Exp}((\tilde{\omega}_k - \mathbf{b}_{gi}) \Delta t) \text{Exp}(-\mathbf{J}_r^k \boldsymbol{\eta}_{gk} \Delta t)] \\ &\doteq \tilde{\Delta \mathbf{R}}_{ij} \prod_{k=i}^{j-1} \text{Exp}(-\Delta \tilde{\mathbf{R}}_{k+1j}^T \mathbf{J}_r^k \boldsymbol{\eta}_{gk} \Delta t) \\ &\doteq \tilde{\Delta \mathbf{R}}_{ij} \text{Exp}(-\delta \phi_{ij}), \end{aligned} \quad (11)$$

where,

$$\begin{aligned} \mathbf{J}_r^k &\doteq \mathbf{J}_r((\hat{\omega}_k - \mathbf{b}_{gi}) \Delta t) \\ \Delta \tilde{\mathbf{R}}_{ij} &\doteq \prod_{k=i}^{j-1} \text{Exp}((\hat{\omega}_k - \mathbf{b}_{gi}) \Delta t). \end{aligned}$$

The equation (11) shows that the IMU preintegral measurement variable $\Delta \mathbf{R}_{ij}$ is composed of two parts, a rotational measurement variable $\Delta \tilde{\mathbf{R}}_{ij}$ without white noise and a corresponding noise $\delta \phi_{ij}$.

Similarly, we obtain the expression of $\Delta \mathbf{v}_{ij}$ and $\Delta \mathbf{p}_{ij}$:

$$\begin{aligned} \Delta \mathbf{v}_{ij} &\approx \sum_{k=i}^{j-1} [\Delta \tilde{\mathbf{R}}_{ik} (\mathbf{I} - (\delta \phi_{ik})^\wedge) (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai}) \Delta t - \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t] \\ &\doteq \Delta \tilde{\mathbf{v}}_{ij} + \sum_{k=i}^{j-1} [\Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai})^\wedge \delta \phi_{ik} \Delta t - \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t] \\ &\doteq \Delta \tilde{\mathbf{v}}_{ij} - \delta \mathbf{v}_{ij} \\ \Delta \mathbf{p}_{ij} &\approx \sum_{k=i}^{j-1} \left[(\Delta \tilde{\mathbf{v}}_{ik} - \delta \mathbf{v}_{ik}) \Delta t + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} (\mathbf{I} - (\delta \phi_{ik})^\wedge) (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai}) \Delta t^2 \right. \\ &\quad \left. - \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t^2 \right] \\ &\doteq \Delta \tilde{\mathbf{p}}_{ij} + \sum_{k=i}^{j-1} \left[-\delta \mathbf{v}_{ik} \Delta t + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai})^\wedge \delta \phi_{ik} \Delta t^2 \right. \\ &\quad \left. - \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t^2 \right] \\ &\doteq \Delta \tilde{\mathbf{p}}_{ij} - \delta \mathbf{p}_{ij}, \end{aligned} \quad (12)$$

where,

$$\begin{aligned} \Delta \tilde{\mathbf{v}}_{ij} &\doteq \sum_{k=i}^{j-1} [\Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai}) \Delta t] \\ \Delta \tilde{\mathbf{p}}_{ij} &\doteq \sum_{k=i}^{j-1} \left[\Delta \tilde{\mathbf{v}}_{ik} \Delta t + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai}) \Delta t^2 \right]. \end{aligned}$$

The error state vector $\boldsymbol{\eta}_{ij}^X$ and error input vector $\boldsymbol{\eta}_j^U$ of the system are defined as:

$$\begin{aligned} \boldsymbol{\eta}_{ij}^X &\doteq [\delta \phi_{ij}^T, \delta \mathbf{v}_{ij}^T, \delta \mathbf{p}_{ij}^T]^T \sim \mathcal{N}(\mathbf{0}_{9 \times 1}, \boldsymbol{\Sigma}_{ij}) \\ \boldsymbol{\eta}_j^U &\doteq [\boldsymbol{\eta}_{gj}^T, \boldsymbol{\eta}_{aj}^T]^T \sim \mathcal{N}(\mathbf{0}_{6 \times 1}, \boldsymbol{\Sigma}_\eta). \end{aligned} \quad (13)$$

The covariance matrices of $\boldsymbol{\eta}_{ij}^X$ and $\boldsymbol{\eta}_j^U$ are $\boldsymbol{\Sigma}_{ij}$ and $\boldsymbol{\Sigma}_\eta$. From equation (11), we have:

$$\text{Exp}(-\delta \phi_{ij}) \doteq \prod_{k=i}^{j-1} \text{Exp}(-\Delta \tilde{\mathbf{R}}_{k+1j}^T \mathbf{J}_r^k \boldsymbol{\eta}_{gk} \Delta t). \quad (14)$$

From equation (12), we get:

$$\begin{aligned} \delta \mathbf{v}_{ij} &\doteq \sum_{k=i}^{j-1} \left[-\Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai})^\wedge \delta \phi_{ik} \Delta t + \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t \right] \\ \delta \mathbf{p}_{ij} &\doteq \sum_{k=i}^{j-1} \left[\delta \mathbf{v}_{ik} \Delta t - \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} (\mathbf{b}_{\hat{a}k} - \mathbf{b}_{ai})^\wedge \delta \phi_{ik} \Delta t^2 \right. \\ &\quad \left. + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} \boldsymbol{\eta}_{ak} \Delta t^2 \right]. \end{aligned} \quad (15)$$

By arranging the equation (14) and the equation (15), we obtain:

$$\begin{aligned} \delta \phi_{ij} &= \Delta \tilde{\mathbf{R}}_{j-1j}^T \delta \phi_{ij-1} + \mathbf{J}_r^{j-1} \boldsymbol{\eta}_{gj-1} \Delta t \\ \delta \mathbf{v}_{ij} &= \delta \mathbf{v}_{ij-1} - \Delta \tilde{\mathbf{R}}_{ij-1} (\mathbf{b}_{\hat{a}j-1} - \mathbf{b}_{ai})^\wedge \delta \phi_{ij-1} \Delta t \\ &\quad + \Delta \tilde{\mathbf{R}}_{ij-1} \boldsymbol{\eta}_{aj-1} \Delta t \\ \delta \mathbf{p}_{ij} &= \delta \mathbf{p}_{ij-1} + \delta \mathbf{v}_{ij-1} \Delta t - \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ij-1} (\mathbf{b}_{\hat{a}j-1} - \mathbf{b}_{ai})^\wedge \delta \phi_{ij-1} \Delta t^2 \\ &\quad + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ij-1} \boldsymbol{\eta}_{aj-1} \Delta t^2. \end{aligned} \quad (16)$$

The equation (16) shows that the error states vector are linear combination of multiple Gaussian noise, and the covariance of the error states vector can be computed from the covariance of the white noise.

We derived the state transition equation of the noise:

$$\eta_{ij}^X = A_{j-1}\eta_{ij-1}^X + B_{j-1}\eta_{j-1}^U, \quad (17)$$

where,

$$A_{j-1} = \begin{bmatrix} \Delta \tilde{R}_{j-1}^T & \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ -\Delta \tilde{R}_{ij-1}(\mathbf{b}_{ai} - \mathbf{b}_{ai})^\wedge \Delta t & \mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ -\frac{1}{2}\Delta \tilde{R}_{ij-1}(\mathbf{b}_{ai} - \mathbf{b}_{ai})^\wedge \Delta t^2 & \Delta t & \mathbf{I}_{3 \times 3} \end{bmatrix}$$

$$B_{j-1} = \begin{bmatrix} \mathbf{J}_r^{-1} \Delta t & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \Delta \tilde{R}_{ij-1} \Delta t \\ \mathbf{0}_{3 \times 3} & \frac{1}{2}\Delta \tilde{R}_{ij-1} \Delta t^2 \end{bmatrix}.$$

The covariance matrix of the error states vector can be computed as follows:

$$\Sigma_{ij} = A_{j-1}\Sigma_{ij-1}A_{j-1}^T + B_{j-1}\Sigma_\eta B_{j-1}^T. \quad (18)$$

3.5 Bias Update

Considering the changing of bias in the optimization process, when a bias update $\mathbf{b} \leftarrow \bar{\mathbf{b}} + \delta \mathbf{b}$ happens, we obtain the approximation of the system states by using the first-order Taylor expansion [10]:

$$\begin{aligned} \Delta \tilde{R}_{ij}(\mathbf{b}_{gi}) &\approx \Delta \tilde{R}_{ij}(\bar{\mathbf{b}}_{gi}) \text{Exp}\left(\frac{\partial \Delta \tilde{R}_{ij}}{\partial \mathbf{b}_g} \delta \mathbf{b}_{gi}\right) \\ \Delta \tilde{\mathbf{v}}_{ij}(\mathbf{b}_{gi}, \mathbf{b}_{ai}) &\approx \Delta \tilde{\mathbf{v}}_{ij}(\bar{\mathbf{b}}_{gi}, \bar{\mathbf{b}}_{ai}) + \frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_g} \Delta \mathbf{b}_{gi} + \frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_a} \Delta \mathbf{b}_{ai} \\ \Delta \tilde{\mathbf{p}}_{ij}(\mathbf{b}_{gi}, \mathbf{b}_{ai}) &\approx \Delta \tilde{\mathbf{p}}_{ij}(\bar{\mathbf{b}}_{gi}, \bar{\mathbf{b}}_{ai}) + \frac{\partial \Delta \tilde{\mathbf{p}}_{ij}}{\partial \mathbf{b}_g} \Delta \mathbf{b}_{gi} + \frac{\partial \Delta \tilde{\mathbf{p}}_{ij}}{\partial \mathbf{b}_a} \Delta \mathbf{b}_{ai}, \end{aligned} \quad (19)$$

where,

$$\begin{aligned} \frac{\partial \Delta \tilde{R}_{ij}}{\partial \mathbf{b}_g} &= -\sum_{k=i}^{j-1} [\Delta \tilde{R}_{k+1,j} \bar{\mathbf{b}}_{gi}^T \mathbf{J}_r^k \Delta t] \\ \frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_a} &= -\sum_{k=i}^{j-1} \Delta \tilde{R}_{ik} \Delta t \\ \frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_g} &= -\sum_{k=i}^{j-1} [\Delta \tilde{R}_{ik}(\mathbf{b}_{ai} - \bar{\mathbf{b}}_{ai})^\wedge \frac{\partial \Delta \tilde{R}_{ik}}{\partial \mathbf{b}_g} \Delta t] \\ \frac{\partial \Delta \tilde{\mathbf{p}}_{ij}}{\partial \mathbf{b}_a} &= \sum_{k=i}^{j-1} \left[\frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_a} \Delta t - \frac{1}{2} \Delta \tilde{R}_{ik} \Delta t^2 \right] \\ \frac{\partial \Delta \tilde{\mathbf{p}}_{ij}}{\partial \mathbf{b}_g} &= \sum_{k=i}^{j-1} \left[\frac{\partial \Delta \tilde{\mathbf{v}}_{ij}}{\partial \mathbf{b}_g} \Delta t - \frac{1}{2} \Delta \tilde{R}_{ik}(\mathbf{b}_{ai} - \bar{\mathbf{b}}_{ai})^\wedge \frac{\partial \Delta \tilde{R}_{ik}}{\partial \mathbf{b}_g} \Delta t^2 \right]. \end{aligned}$$

4 Visual-Inertial Odometry

In our visual-inertial odometry, we track the camera data using LK Optical flow algorithm [11]. Then, EPnP method is used to estimate the camera motion. At the same time, the IMU measurement are dealt with by the IMU preintegration algorithm. After bundle adjustment optimization, we get the final optimized pose. The flow chart of the scheme is shown in Fig. 1.

5 Experiments and Analysis

To validate the proposed preintegration algorithm, we implemented the algorithm on the Ubuntu system using C++.

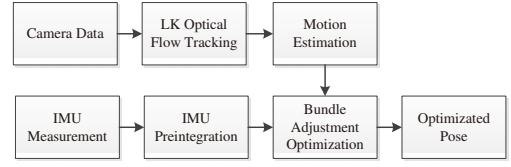


Fig. 1: The flow chart of our scheme.

5.1 Odometry Experiment

We tested our IMU preintegration method on EuRoC Visual-Inertial datasets, which collected on-board a Micro Aerial Vehicle (MAV). The EuRoC datasets contain stereo images (20 Hz), synchronized IMU measurements (200 Hz), and accurate motion and structure ground-truth. Large IMU bias and illumination changes are observed in these datasets.

In our experiments, we compare our algorithm with famous Okvis, a state-of-the-art VIO algorithm. We use two sequences in EuRoC, the MH_05_difficult and V1_03_difficult. The experimental results are showed below.

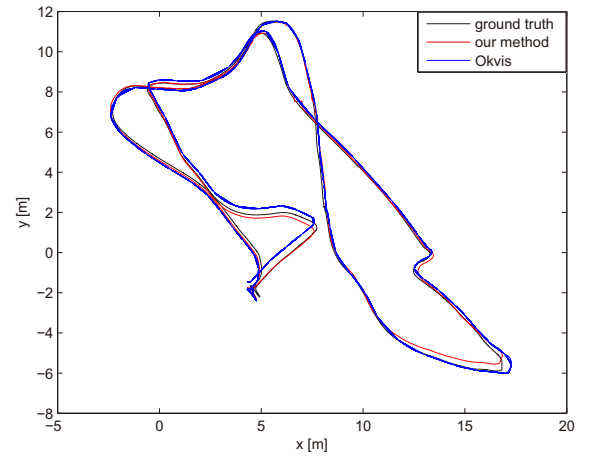


Fig. 2: Trajectory in MH_05_difficult.

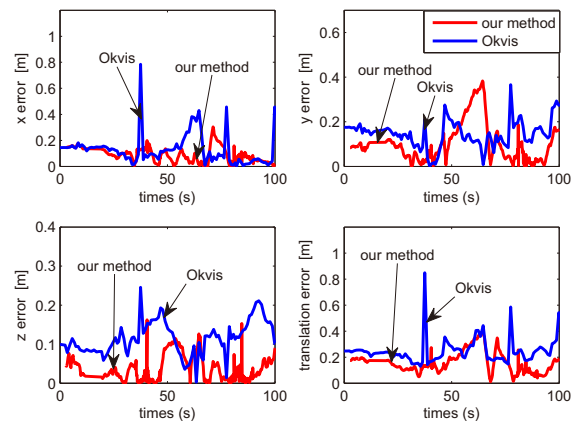


Fig. 3: Translation error in MH_05_difficult.

Firstly, we tested our algorithm on dataset MH_05_difficult with the Okvis algorithm. The camera trajectory is shown in Fig. 2. The trajectory curve shows that both the our method and the Okvis algorithm can track the motion of the camera in real time when the camera moving

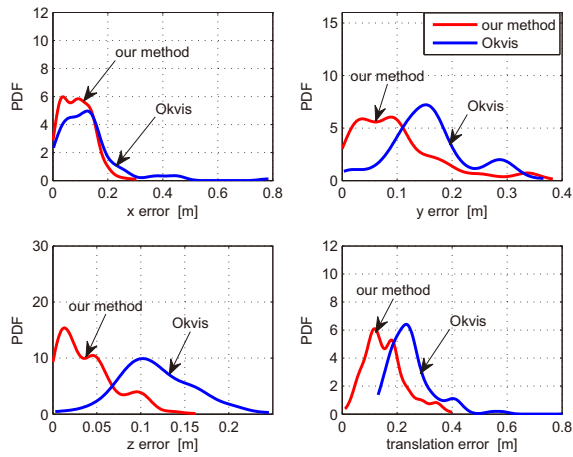


Fig. 4: Translation error PDF in MH_05_difficult.

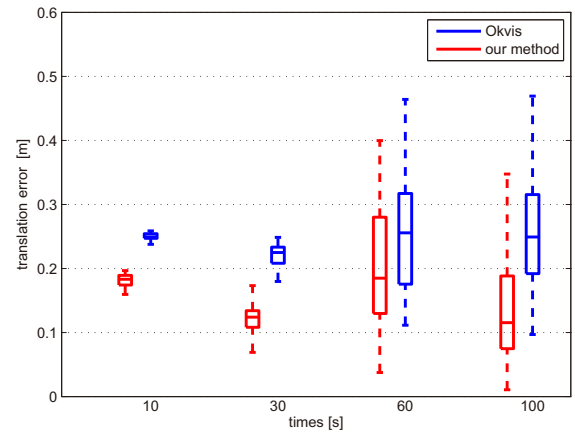


Fig. 7: Error boxes in different times in MH_05_difficult.

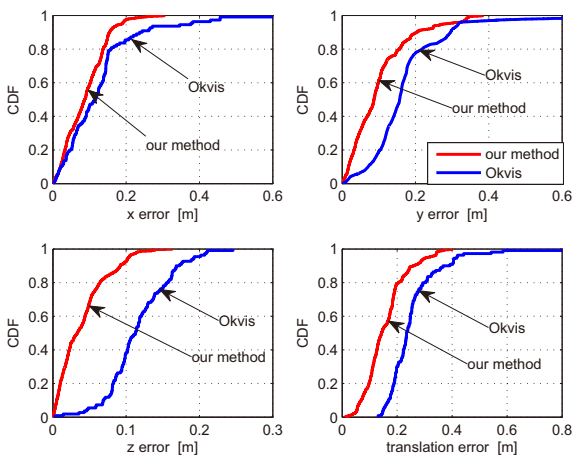


Fig. 5: Translation error CDF in MH_05_difficult.

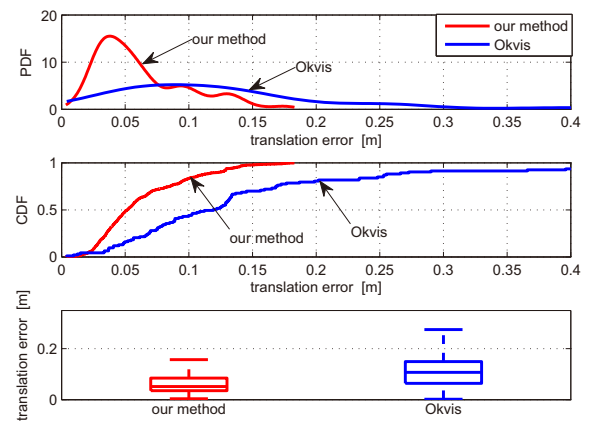


Fig. 8: Translation error distribution in V1_03_difficult.

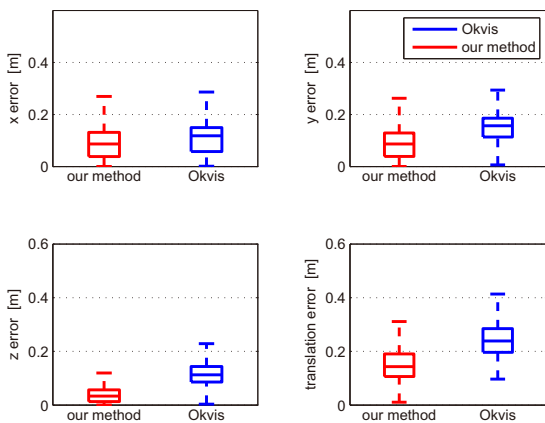


Fig. 6: Translation error box in MH_05_difficult.

in a range of about $20\text{ m} \times 20\text{ m}$.

The translation error curves are shown in Fig. 3. The curves show that our algorithm limits the error on the x axis and the y axis to 0.4 m, the error on the z axis to 0.15 m, and the total translation error to 0.4 m range. The corresponding results of the Okvis algorithm are 0.8 m, 0.4 m, 0.25 m and

0.9 m.

The comparisons of translation error probability density function (PDF) and cumulative distribution function (CDF) between our algorithm and the Okvis algorithm are shown in Fig. 4 and Fig. 5. The CDF value of a point indicates the probability that the error is less than this point. The CDF graph shows the probability of the tracking error less than 0.3 m is 95% and the probability of the error less than 0.18 m is 67% in our algorithm. The corresponding results of the Okvis algorithm are 80%(0.3 m) and 15% (0.18 m).

The translation error box plots are shown in Fig. 6. It can be seen from Fig. 6 the median tracking error of our algorithm is 0.14 m, the lower quartile error is 0.11 m and the upper quartile error is 0.19 m. For the Okvis algorithm, the corresponding values are 0.24 m, 0.2 m and 0.28 m. The translation error box plots at different times are shown in Fig. 7. We divided the trajectory into four sections, $[0\ 10\text{ s}]$, $[10\text{ s}\ 30\text{ s}]$, $[30\text{ s}\ 60\text{ s}]$, $[60\text{ s}\ 100\text{ s}]$, respectively. The error distributions of our algorithm and the Okvis algorithm are computed for each sub-trajectory, and four groups of box plots are obtained. As shown in Fig. 7, with the growth of time, the error of our algorithm can be maintained within the range of 0.4 m, and the median error within 0.2 m.

The Table 1 shows the other tracking evaluation compar-

ison between our algorithm and the Okvis algorithm. In the table, ATE stands for absolute trajectory error and it measures the difference between points of the true and the estimated trajectory [12]. PRE means relative pose error and it measures the error in the relative motion between pairs of timestamps [12]. Our algorithm has an ATE of 0.17 m, an RPE of 0.23 m and an average rotation error of 0.88° . The corresponding values of Okvis algorithm are 0.37 m, 0.49 m and 1.67° .

Table 1: Experiment evaluation in MH_05_difficult.

Evaluation Item	Our Method	Okvis Algo.
ATE (m)	0.1737	0.3797
RPE (m)	0.2284	0.4986
Rotational_error.max ($^\circ$)	4.37	33.56
Rotational_error.mean ($^\circ$)	0.88	1.67
Rotational_error.std ($^\circ$)	0.58	2.58

In the experiment on the dataset V1_03_difficult, the translation error distribution as shown in Fig. 8 and the evaluation of tracking is shown in Table 2.

Table 2: Experiment evaluation in V1_03_difficult.

Evaluation Item	Our Method	Okvis Algo.
ATE (m)	0.0723	0.1933
RPE (m)	0.1132	0.2691
Rotational_error.max ($^\circ$)	7.97	65.30
Rotational_error.mean ($^\circ$)	2.26	4.18
Rotational_error.std ($^\circ$)	1.44	8.35

The above experimental results show that both our algorithm and the Okvis algorithm can well track the pose of the camera under the EuRoC dataset. Compared with the Okvis algorithm, our algorithm has a significant improvement in tracking accuracy.

5.2 Noise Propagation Experiment

We test our noise propagation method on MH_05_difficult sequence, EuRoC datasets. The translation error result is showed in Fig. 9.

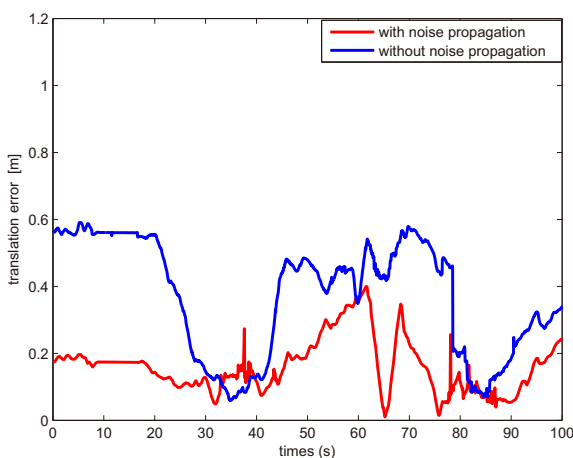


Fig. 9: Translation error in noise propagation experiment.

In our experiment, the odometry without noise propagation has an ATE of 0.41 m, an RPE of 0.49 m and a median

tracking translation error of 0.42 m. The corresponding values of the noise propagation method are 0.17 m, 0.23 m and 0.14 m.

The experimental results show that the noise propagation method algorithm has a significant improvement in tracking accuracy.

6 Conclusion

In this paper, we propose an IMU preintegration method for VIO pose estimation. By preintegrating IMU measurement on manifold, the proposed approach avoid effectively the re-integration process. The influence of the noise on pose estimation can be dramatically reduced because our method studied the propagation characteristic of noise and derived the state transition equation of the noise. Finally, we validated our algorithm on two public datasets and achieved good tracking accuracy.

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